MANAV RACHNA UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & TECHNOLOGY

"Mid Semester Examination, March-2026"

SEMESTER	VI	DATE OF EXAM	18.03.2026	CREDITS	3
COURSE CODE	CSH414B-T	SESSION	III	PROGRAM	B.TECH CSE (CORE + AIML/CSTI/FSD
COURSE NAME	INFORMATION RETRIEVAL	MAX. MARKS	40	TIME DURATION	90 MINUTES
Students will be assessed in the *Mid-Sem Exam* of this course on the following:					
WK3-Engineering Fundamentals		CO1	SK1-Problem Solving		CO1, CO2
WK6-Engineering Practice		CO1, CO2			

Note:- All questions are compulsory.

PART – A

Q. NO.	QUESTIONS	MARKS	CO	BLOOMS LEVEL
1	Translate the following into First Order Logic:			
1.(a)	Ajay likes all kinds of bike.	2	CO1	BT2
1.(b)	Anything anyone eats and not killed is food.	2	CO1	BT2
2.	Differentiate Forward Chaining and Backward Chaining with one example of each.	2	CO1	BT2
3.	Construct the truth table for the expression: $(p \wedge q) \rightarrow p(p \wedge q) \rightarrow p(p \wedge q) \rightarrow p$ State whether it is a tautology.	4	CO1	BT3
4.	Draw a semantic network to represent: • A sparrow is a bird • A bird is an animal • Birds fly • Tweety is a sparrow	3	CO1	BT3
5.	What is Information Retrieval? Why retrieving information is difficult?	2	CO1	BT2

PART – B

Q.NO.	QUESTIONS	MARKS	CO	BLOOMS LEVEL
6.	Given the following inverted index: Term: apple — Posting List: {1, 2, 4, 6}	5	CO2	BT4

	banana {2, 4, 5} mango {1, 3, 4} Analyze and solve the Boolean query: (a) apple AND banana (b) apple OR mango (c) apple AND NOT banana Show all intermediate steps.			
7.	a) Explain Positional Indexing with an example. b) For the following documents, answer the phrase query "data mining": D1: data mining techniques D2: mining data analysis Show how positional index processes the query.	2+3=5	CO2	BT3
8.	Consider the following axioms: a. All people who are graduating are happy. b. All happy people smile. c. Someone is graduating. Prove by resolution that: Someone is smiling. Show all the Steps of resolution • Conversion of facts into first-order logic. • Convert FOL statements into CNF • Negate the statement which needs to prove (proof by contradiction) • Draw resolution graph (unification).	3+2+2+3 = 10	CO2	BT3
9.	a) Compare Boolean Retrieval Model and Vector Space Model. b) Why does Boolean Retrieval fail to provide ranked results? How does VSM overcome this limitation?	3+2=5	CO2	BT2